

### 5.1 Classwork: Introduction to Kinematics — Solutions

1. A runner jogs 400 m east and then 150 m west.
  - (a) Find the total **distance** the runner travels. [1]
  - (b) Find the runner's **displacement** from the starting point. State the direction. [2]
  - (c) Explain why these two answers are different. [1]

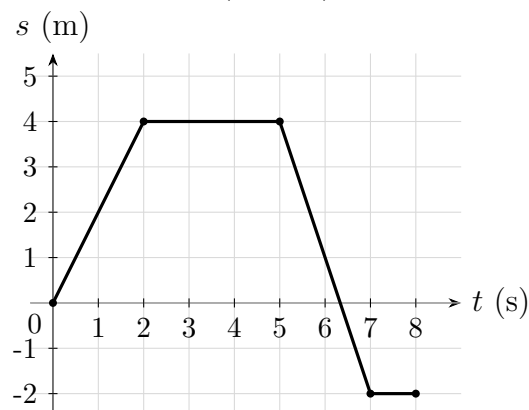
#### Solutions

(a) Distance =  $400 + 150 = \mathbf{550}$  m

(b) Displacement =  $400 - 150 = \mathbf{250}$  m east

(c) Distance counts the total path length (scalar); displacement measures the net change in position from start to finish (vector). The runner reversed direction, so part of the path cancels in displacement but not in distance.

2. The graph below shows the position  $s$  (metres) of a particle at time  $t$  (seconds).



- (a) Find the velocity of the particle during  $0 \leq t \leq 2$ . [2]
- (b) Describe the motion of the particle during  $2 \leq t \leq 5$ . [1]
- (c) Find the velocity during  $5 \leq t \leq 7$ . [2]
- (d) Find the total **distance** travelled from  $t = 0$  to  $t = 7$ . [2]
- (e) Find the **displacement** from  $t = 0$  to  $t = 7$ . [1]

**Solutions**

(a)  $v = \frac{\Delta s}{\Delta t} = \frac{4 - 0}{2 - 0} = \mathbf{2 \text{ m s}^{-1}}$

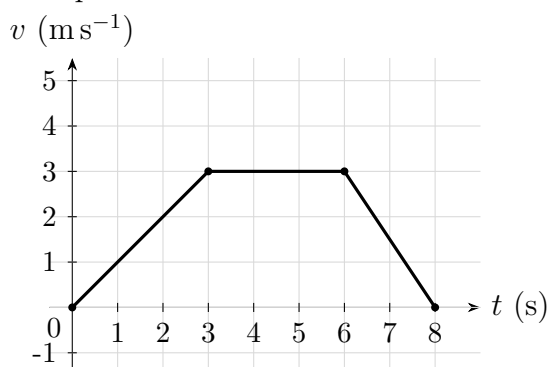
(b) The particle is stationary (displacement is constant at 4 m).

(c)  $v = \frac{-2 - 4}{7 - 5} = \frac{-6}{2} = \mathbf{-3 \text{ m s}^{-1}}$

(d) Distance =  $|4 - 0| + |4 - 4| + |-2 - 4| = 4 + 0 + 6 = \mathbf{10 \text{ m}}$

(e) Displacement =  $s(7) - s(0) = -2 - 0 = \mathbf{-2 \text{ m}}$

3. The velocity  $v$  ( $\text{m s}^{-1}$ ) of a particle is shown below.



- (a) Find the acceleration during  $0 \leq t \leq 3$ . [2]
- (b) State the acceleration during  $3 \leq t \leq 6$ . [1]
- (c) Find the acceleration during  $6 \leq t \leq 8$ . [2]
- (d) Is the particle slowing down during  $6 \leq t \leq 8$ ? Explain your answer. [1]

### Solutions

(a)  $a = \frac{\Delta v}{\Delta t} = \frac{3 - 0}{3 - 0} = \mathbf{1 \text{ m s}^{-2}}$

(b)  $a = \mathbf{0 \text{ m s}^{-2}}$  (constant velocity)

(c)  $a = \frac{0 - 3}{8 - 6} = \frac{-3}{2} = \mathbf{-1.5 \text{ m s}^{-2}}$

(d) Yes. The velocity is positive but decreasing toward zero, so the particle is moving forward but slowing down (decelerating).

- 4. Scalar or vector. [4]
- 5. Negative velocity at  $t = 4$ .
- 6. Position table:  $s = 2, 5, 8, 11, 14$ .

7. Cyclist from rest,  $a = 2 \text{ m s}^{-2}$  for 5 seconds.

### Solutions to Problems 4–7

**Problem 4:** (a) Scalar (speed, no direction given)

(b) Vector (displacement, direction north)

(c) Vector (velocity, has sign)

(d) Scalar (temperature)

**Problem 5:**

(a) The negative sign means the particle is moving in the negative direction (to the left).

(b) Speed  $= |v| = |-3| = \mathbf{3 \text{ m s}^{-1}}$

**Problem 6:**

(a)  $v = \frac{\Delta s}{\Delta t} = \frac{5 - 2}{1 - 0} = \mathbf{3 \text{ m s}^{-1}}$  (constant, same for every interval)

(b)  $a = \mathbf{0 \text{ m s}^{-2}}$  (velocity is constant)

(c) The particle moves at a constant velocity of  $3 \text{ m s}^{-1}$  in the positive direction (uniform motion, no acceleration).

**Problem 7:**

(a)  $v = u + at = 0 + 2 \times 5 = \mathbf{10 \text{ m s}^{-1}}$

(b) Straight line from  $(0, 0)$  to  $(5, 10)$  on the  $v$ - $t$  graph.